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January 11, 1974
SUPERSEDING
Fed. Spec. UU-E-522D
May 19, 1969

FEDERAL SPECIFICATION

ENVELOPE, MAILING

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers mailing envelopes for the departments and independent establishments of the Government.

1.2 Classification.

1.2.1 Types, classes, and forms. Mailing envelopes covered by this specification shall be the following types, classes and forms:

Type I - Light brown color.
Type II - White, regular.
Type III - White, airmail.
Type IV - Colored.
Type V - White, bond.
Type VI - White, with opaquing design.
Class 1 - With windows.
Class 2 - Without windows.
Form 1 - Diagonal seam, high cut.
Form 2 - Diagonal seam, web feed.
Form 3 - Side seam.
Form 4 - Center seam.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

GG-M-571 - Moistener, Paper Sealing (Pencil Type for Envelopes).
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Setup.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

Postal Service Manual

Code of Federal Regulations - CFR 101-11.603.9

(The Code of Federal Regulations (CFR) and the Postal Service Manual are for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Assocation, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Technical Manual of the American Association of Textile Chemists and Colorists.

(Application for copies should be addressed to the Executive Secretary, AATCC National Headquarters, Lowell, MA 01854, or Howes Publishing Corporation, 44 East 23rd St., New York, NY 10010.)

Technical Association of the Pulp and Paper Industry (TAPPI) Standards:

T 401 - Fiber Analysis of Paper and Paperboard.
T 402 - Standard Conditioning and Testing Atmospheres for Paper, Board, Pulp Handsheets and Related Products.
T 403 - Bursting Strength of Paper.
T 410 - Weight per Unit Area (Basis Weight or Substance) of Paper and Paperboard.
T 425 - Opacity of Paper.
T 479 - Smoothness of Printing Paper (Bekk).
T 633 - Preparation of Magnesium Oxide Standard for Spectral Reflectivity.

(Application for copies should be addressed to the Technical Association of the Pulp and Paper Industry, One Dunwoody Park, Atlanta, GA 30341.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material. Mailing envelopes shall be made from fibrous, cellulosic material which may include reclaimed fibers and shall conform to the requirements of this specification.

3.1.1 Basis weight tolerance. A tolerance of plus or minus five percent shall be allowed in the basis weight of the paper used in the envelopes.

3.2 Writing quality. All paper used to fabricate envelopes shall be sufficiently sized to permit the acceptance of ink, without feathering, spreading, or penetration to the reverse side, when tested as specified in 4.2. When the paper is surface sized, the sizing shall be starch or animal glue.

3.3 Finish (smoothness). The finish (smoothness) of the paper used to fabricate envelopes shall average not less than 5 seconds nor more than 150 seconds when tested as specified in 4.2.1. Except as specified in 3.2 the use of chemical finishes or treatments is prohibited.

3.4 Size. The envelope dimensions shall be as specified (see 6.2), with a tolerance of ± 1/16 inch for each dimension.

3.5 Type I, envelopes, light brown.

3.5.1 Color. Envelopes shall be buff, manila or light brown. The requirements of the specification to apply in all other respects.

3.5.2 Basis weight and bursting strength shall comply with the requirements of table I.

TABLE I. Basis weight and bursting strength for envelopes, light brown

Basis weight, 17 by 22 - 500 sheets	Bursting strength, average, not less than
Pounds	Pounds per square inch (psi)
20	30
24	38
28	44
32	50
40	60

3.6 Types II and III, envelopes, white regular and white air mail.

3.6.1 Physical and optical characteristics shall comply with the requirements of table II.

TABLE II. Physical and optical characteristics for envelopes, white regular and white air-mail

Basis weight, 17 by 22 500 sheets	Bursting strength, average, not less than	Opacity (contrast ratio)
Pounds	Pounds per square inch, (psi)	Percent
20	20	86
24	24	89

3.7 Type IV, envelopes, colored.

3.7.1 Colors. Envelopes shall be blue, light green, pink, or goldenrod, as specified (see 6.2).

3.7.1.1 Blue and light green envelopes shall be equal to or lighter in hue than the standard sample as furnished by the contracting officer (see 6.3).

3.7.1.2 Pink and goldenrod envelopes shall be in accordance with commercial practice.

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3.7.2 Basis weight and bursting strength shall comply with the requirements of table III.

TABLE III. Basis weight and bursting strength for envelopes, colored

Basis weight, 17 by 22 - 500 sheets	Bursting strength, average, not less than
Pounds	Pounds per square inch, (psi)
20	20

3.8 Type V, envelopes, bond, white.

3.8.1 Stock. The paper shall be made from not less than 25 percent cotton or linen textile mill wastes, linters or rags; the remainder free from unbleached chemical wood or groundwood pulp.

3.8.2 Basis weight and bursting strength shall comply with the requirements of table IV.

TABLE IV. Basis weight and bursting strength for envelopes, white bond

Basis weight, 17 by 22 - 500 sheets	Bursting strength, average, not less than
Pounds	Pounds per square inch, (psi)
24	38

3.9 Type VI, envelopes, white, with opaquing design printed inside.

3.9.1 Opaquing design. The opaquing design shall be printed in black or blue ink on the inside of the envelope, and shall give the envelope sufficient opacity to prevent the distinguishing of any letters or figure when tested as specified in 4.4.2. Opaquing designs should be as uniform as possible and should avoid the appearance of lines of type parallel to the long edge of the envelope.

3.9.2 Basis weight and bursting strength shall comply with the requirements of table V.

TABLE V. Basis weight and bursting strength for envelopes, white with opaquing design printed inside

Basis weight, 17 by 22 - 500 sheets	Bursting strength, average, not less than
Pounds	Pounds per square inch, (psi)
20	20

3.10 Class 1, window envelopes. Envelope windows shall be covered with a transparent material, conforming to the requirements set forth below, glued close to the edge of the cut-out; alternatively, the window area may be an integral part of the envelope face provided that it conforms to the requirements for window material in 3.10.1.

3.10.1 Window material.

3.10.1.1 Reflectance. The window material shall have a reflectance not less than 70 percent when measured at 400 millimicrons with respect to magnesium oxide, when tested as specified in 4.2.

3.10.1.2 Opacity. The window material shall have an opacity of not more than 25 percent, expressed as the ratio of the reflectance at 400 millimicrons with a black backing to the reflectance at 400 millimicrons with a magnesium oxide backing, when tested as specified in 4.2.

3.10.2 Window location. Address windows shall be rectangular with round or slightly rounded corners (see 6.2). The window size and location shall be such as to encompass an address located within an address read zone and provide 1/4 inch clearance between the address and the left, right, and bottom edges of the window. The window itself may extend outside of the read zone. The address read zone is a

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rectangular area 2-1/2 by 8 inches, parallel to the long edge of the envelope, starting one inch from the left envelope edge and extending 8 inches to the right (or to the right edge of the envelope, if it is less than 9 inches long) and starting 1/2 inch above the bottom envelope edge and extending upward for 2-1/2 inches. The size and location of the window, within the above limitations shall be as specified (see 6.2).

3.11 Adhesive. The adhesives used in accordance with this specification shall be nontoxic, free from offensive odor, and free from ingredients which would tend to discolor the paper when tested as specified in 4.2.2.

3.11.1 Seam adhesive. The seam adhesive shall be of such quantity and consistency as to provide firmly bonded seams without unreasonable curling of the envelope. Unreasonable curling is that which would interfere with the intended use or with the passage of the envelope through mail handling equipment.

3.11.2 Flap adhesive. The flap adhesive (regular or temperature-humidity resistant) shall show pronounced tackiness when moistened and shall permit the flap to be quickly and securely sealed by ordinary pressure. When opened 15 seconds, -0 or +5 seconds after sealing, the flaps shall pull fibers from the body of the envelope along at least 75 percent of the gummed length when tested as specified in 4.3.2.

3.11.2.1 Regular adhesive. Unless otherwise specified, the flap adhesive shall not cause premature adhesion of the sealing flaps when envelopes are used or stored under nontropical conditions.

3.11.2.2 Temperature and humidity resistant flap adhesive. When specified (see 6.2), a temperature and humidity resistant flap adhesive shall be used. The flap adhesive shall permit envelopes to be stored under a weight of 10 grams per square inch of envelope area for 24 hours at 100° ± 2°F. and 80 ± 2 percent relative humidity, without adhesion of the flap to the envelope back when tested as specified in 4.3.2.3.

3.12 Construction. Envelopes in accordance with this specification shall be of diagonal seam (high cut or web feed), side seam, or center seam construction, as specified (see 6.2).

3.12.1 Form 1, diagonal seam, high cut. Constructional details, as identified by the letters shown on Illustration No. 1, shall be as shown below.

A - The sealing flap shall overlap the side flaps not less than 3/8 inch along area where adhesive is applied to the sealing flap.

AA - The sealing flap shall overlap the bottom flap for a distance not less than 10 percent of the envelope width, and in no case shall this distance be less than 5/8 inch.

B - The opening under the sealing flap shall not be more than one-half of the flap, identified as C.

D - The width of each side flap shall be at least 30 percent of the length of the envelope, but in no case shall this distance be less than 2-1/2 inches.

E - The width of the overlap of the diagonal sealed seam shall be not less than 7/16 inch.

F - The width of the adhesive on diagonal sealed seams shall be not less than one-third the width of the seam (no minus tolerance permitted) and shall extend to within one-half inch of the bottom corners.

3.12.2 Form 2, diagonal seam, web feed. Constructional details, as identified by the letters shown on Illustration No. 2, shall be as shown below.

A - The sealing flap shall overlap the side flaps for a distance not less than 5/8 inch along area where adhesive is applied to sealing flap.

AA - The sealing flap shall overlap the lower flap not less than 3/8 inch.

B - The opening under the sealing flap shall be not greater than 70 percent of the flap, identified as C.

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C - The distance from the fold to the apex of the flap shall not exceed 2-1/2 inches.

D - The side flaps shall be of sufficient width to abut against each other.

E - The width or overlap of the diagonal sealed seams shall be not less than 7/16 inch.

F - The width of the adhesive on diagonal sealed seams shall be not less than one-third of the width of the seam (no minus tolerance permitted) and shall extend to within one-half inch of the bottom corners.

3.12.3 Form 3, side seam. Constructional details as identified by the letters shown on Illustration No. 3, shall be as shown below.

A - The sealing flap shall overlap the lower flap for a distance not less than 5/8 inch along area where adhesive is applied to the sealing flap.

B - The opening under the sealing flap shall be not greater than one-half of the flap, identified as C.

NOTE: Although Illustration No. 3 shows a V-shaped sealing flap, a square-cut, sealing flap as shown in Illustration No. 4, is acceptable. However, the requirements of A and B above shall be met.

D - The width of each side flap shall be not less than 7/8 inch.

E - The width or overlap of the side sealed seams shall be not less than 7/16 inch.

F - The width of the adhesive on side sealed seams shall be not less than one-third the width of the seam and shall come to within not more than 1/8 inch of the outermost edge of the back flap. Insofar as practicable, the adhesive should extend the full length of the side seam and in no case shall the adhesive end be at a point more than 1/4 inch above the bottom edge of the seam.

3.12.4 Form 4, center seam. Center seam envelopes shall be light brown envelopes with one open end forming the sealing flap. Constructional details as identified by the letters shown on Illustration No. 4, shall be as shown below.

A - The sealing flap shall overlap the lower flap for a distance not less than 1 inch along area where adhesive is applied to the sealing flap.

B - The opening under the sealing flap shall be not greater than one-half of the flap identified as C.

D - The width of each side flap shall be at least 54 percent of the length of the envelope.

E - The width of the overlap of the two side flaps shall be not less than 7/16 inch.

F - The width of the adhesive on the side flap sealed seam shall be not less than one-third the width of the seam (no minus tolerance permitted).

J - The width of the overlap of the bottom flap shall be not less than 10 percent of the width of the envelope. In no case shall this overlap be less than 7/16 inch.

K - The width of the adhesive on the bottom flap sealed seam shall be not less than one-third the width of the seam (no minus tolerance permitted) and shall extend to within one-half inch of the sides.

3.12.5 Sealing flaps, all constructions. The depth of the sealing flap (identified as C on all illustrations) shall be in accordance with standard commerical practice except as otherwise specified herein.

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3.12.5.1 For envelopes intended for use in atuomatic inserting and sealing equipment the flap depth shall be such as to insure satisfactory operation in the machine for which it is intended (see 6.2).

3.12.5.2 When specified (see 6.2), envelopes shall be supplied with ungummed sealing flaps.

3.12.5.3 When a temperature and humidity resistant flap adhesive is specified (see 3.11.2.2), the sealing flap shall be ungummed at the point where they cross the back seams (split gummed). Unless otherwise specified (see 6.2) split gumming may be furnished on all other envelopes, at the option of the contractor.

3.12.5.4 Except as provided in 3.12.5.3 the flap adhesive shall be applied along the length of the flap to within not more than 1/2 inch of each end of the flap (identified as "H" on all illustrations). The adhesive line may be of tapered (see "G" Illustrations Nos. 1 & 2) or square cut (see "G" Illustration No. 3) construction.

3.12.5.5 The width of the sealing flap adhesive for all envelopes with a front area of 21 square inches or less shall be not less than 7/16 inches, or shall be 3/4 inch at the apex and taper to the edge. The width of the adhesive on envelopes of larger size shall be proportionately increased with a minimum of one inch width for envelopes of a front area of 120 square inches or more (see "G", on all illustrations).

3.13 Printing of indicia envelopes.

3.13.1 Envelopes may be printed on the face and back. Unless otherwise authorized (see 3.13.3), face printing shall be confined to the upper right and left hand corners of the envelope and shall be in accordance with Chapter 1, Part 137 of the Postal Service Manual. The eagle symbol forming part of the indicia shall be located 3/8 inch, plus or minus 1/16 inch, from the top and right edges of the envelope. The upper right corner printing as required by the Postal Service Manual shall be centered and appear to the left of the eagle symbol (see example below).

3.13.2 Airmail envelopes. Airmail envelopes shall be printed with a border containing alternating blue and red parallelograms with intervening spaces between the two. The size of the parallelogram shall be approximately 3/8 inch wide by 3/8 inch long and shall form a border on the face of the envelope not greater than 1/4 inch in width. The face printing shall be as specified in 3.13.1, except that the eagle indicia shall be located 1/2 inch, plus or minus 1/16 inch from the top and right edges of the envelope and the word "airmail" shall appear under the upper right hand corner printing.

3.13.3 The printing of borders, markings, slogans, or designs on envelopes may be included on special justification as provided under United States Code, Title 41, subchapter B - Archives and Records, Part 101-11.603.9.

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3.13.4 Color of ink. Except as noted in 3.13.2 and 3.13.3, printing shall be black, blue-black, or blue.

3.14 Color. The color of the envelopes shall be as specified in 3.5.1, 3.6, 3.7.1, 3.8, and 3.9. The envelope stock shall have no optical brighteners added to the pulp or paper during manufacture. The envelopes shall show fluorescence of not more than 2 percent when tested as specified in 4.4.6.

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3.15 Workmanship. Mailing envelopes, in accordance with this specification, shall be uniformly made and shall be free from defects with respect to color, cleanliness, pin holes, breaks, tears, cuts, sharp creases, wrinkles, warpage, free from spots and other imperfections which would impair the serviceability of the envelope. Envelopes shall be cut sufficiently high to prevent adhesive from coming in contact with contents. The sealed flaps or seams shall be sufficiently wide to insure strength and sealed with such quantity and quality of adhesive as will effect perfect and permanent adhesion of flaps and seams. The number of defects shall not exceed the acceptable quality level in section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.1.2 Inspection of components and materials. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials used were manufactured, examined, and tested in accordance with this specification or referenced subsidiary specifications and standards.

4.1.2.1 Component testing. Each lot of paper or, as applicable, window covering material used in the fabrication of envelopes shall be tested in accordance with the applicable test methods specified hereinafter, to determine compliance with the requirements of section 3. Each inspection lot shall consist of a component of a single type or grade manufactured by the same source, under essentially the same conditions and used in the production of the quantity of the end item submitted for inspection at one time.

4.1.2.2 Evidence of compliance. In lieu of using his own facilities, the contractor may submit a report of analysis of component materials from a laboratory acceptable to the Government, whether an independent test facility or the laboratory of the raw material supplier, as evidence of compliance with the specification requirements. The report of analysis shall include, in addition to the actual test data, the manufacturer's name, name and address of the testing laboratory, date of test, and quantity of material represented by the test report. Where certificates of compliance or reports of analysis are submitted, the Government reserves the right to check test such items to determine the validity of the certification or analysis.

4.2 Testing of components.

4.2.1 Paper and window material. Component materials shall be tested for the applicable characteristics specified in table VI. The test shall be made in accordance with the indicated TAPPI methods or as referenced in the table. The lot size for the purpose of determining the sample size shall be expressed in pounds of paper, or as applicable, window covering material, used in the manufacture of the envelopes. The sample unit shall consist of not less than 2,000 square inches of base paper or window material. The sample size shall be drawn in accordance with level S-1 of MIL-STD-105. Test results shall include all values upon which the results are based and there shall be no failures to meet the composite, sample unit, or lot average requirements. The composite sample shall consist of approximately equal amounts drawn from all the sample units combined into a single sample. Except where otherwise specified, all samples shall be conditioned and tested under uniform atmospheric conditions as specified in TAPPI method T 402.

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TABLE VI. Instructions for testing components

Characteristic	Spec Rqmt	Ref Test Meth	Rqmts to Sample unit	Appli to Lot average	Results reported as Pass or Fail	Results reported as Numer to Nearest
Atmospheric conditions	4.2	T 402 1/	All testing			
Paper						
Writing quality	3.2	4.4.4	X		X	
Smoothness (Bekk)	3.3	T 479 1/	X			second
Fiber identification	3.8.1	T 401 1/	Composite			5 percent
Basis weight	Tables I, II, III, IV, V	T 410 1/		X		0.1 pound
Bursting strength	Tables I, II, III, IV, V	T 403 1/		X		1 psi
Opacity	Table II	T 425 1/		X		0.1 percent
Fluorescence	3.14	4.4.6	X		X	
Window material						
Reflectance	3.10.1.1	4.4.5.3	X			0.5 percent
Opacity	3.10.1.2	4.4.5.4	X			0.5 percent

1/ Refers to applicable TAPPI method.

4.2.2 Adhesive. The supplier shall submit a certificate of compliance stating that the adhesive conforms to the requirements in 3.11.

4.3 Inspection of the end item.

4.3.1 Examination of the end item. The end item shall be examined for the defects set forth in the applicable subparagraphs at the inspection levels and acceptable quality levels (AQLs) in 4.3.1.4. Random samples shall be drawn from each lot for examination of visual, dimensional, and preparation for delivery defects. The lot size shall be expressed in units of envelopes for examinations in 4.3.1.1 and 4.3.1.2 and in units of shipping containers in the end item inspection lot for examination in 4.3.1.3.

4.3.1.1 Examination for visual defects. The sample unit for this examination shall be one envelope. Both sides of the material shall be examined. No more than one sample unit shall be selected from any one package and when practicable, no more than three sample units from any one shipping container. A defect of any type shall be counted only once in a sample unit.

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Examine	Defect
Color	Not color specified. Does not compare with Government standard sample, when required.
Appearance	Presence of dirt, grease, foreign matter, or slime spots. Wrinkle, folds, holes, tears, cuts, warpage or sharp creases. Surfaces not clean; not free from paper dust.
Workmanship	Adhesive adheres to contents. Width of sealed flaps or seams not as specified. Flaps not extended, when specified. Quantity and quality of adhesive does not provide permanent adhesion of flaps and seams; flap adhesive not as specified or required. Length of flaps not as specified. Window opening not as specified. Windows not provided with covering; colored covering. Window covering not glued properly, does not lie flat. Window-corners not as specified. Printing not as specified in size or type, or of proper color; indicium missing; agency address missing; Airmail border missing; of incorrect size or sequence; word "airmail" missing.

4.3.1.2 Examination for dimensions. The sample unit for this examination shall be one envelope. No more than one sample unit shall be selected from the same unit package.

Examine	Defect
Size	Varies from dimensions specified by more than ± 1/16 inch. Not size specified.
Window	Window size not as specified.
Airmail border	More than 1/4 inch in width.

4.3.1.3 Examination of preparation for delivery. An examination shall be made to determine that markings, materials, and workmanship comply with the requirements of section 5. The sample unit shall be one fully packed shipping container. The shipping containers shall be examined for closure defects.

Examine	Defect
Marking (exterior and interior)	Incorrect; incomplete; illegible; omitted; of improper size, location, sequence, or method of application.
Material	Any nonconforming component; component missing, damaged or otherwise defective.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps; loose or inadequate sealing, strapping, or stapling.
Contents	Number of intermediate packages per shipping container is less than specified.

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4.3.1.3.1 Count. An examination shall be made to determine the average count of envelopes per package. The lot size shall be expressed in units of intermediate packages. The sample unit shall be one intermediate package and shall be from the shipping containers selected for examination in 4.3.1.3.

Examine	Defect
Count	Average count per package for all packages examined is less than specified.
Count marker, as specified	Separation slips or turned up flap missing, when specified. More or less than 50 envelopes between markers.

4.3.1.4 Inspection levels and acceptable quality levels for examination. The inspection levels for determining the sample size and AQLs, expressed in defects per 100 envelopes, 100 shipping containers, or 100 intermediate packages, as applicable, shall be as follows:

Examination paragraph	Inspection levels	AQLs
4.3.1.1	I	6.5
4.3.1.2	S-2	2.5
4.3.1.3	S-2	2.5
4.3.1.3.1	S-2	2.5

4.3.2 Testing of the end item.

4.3.2.1 All envelopes. All envelopes shall be tested as specified in 4.4.1 for compliance with the requirements in 3.11.2. Ten envelopes for test shall be randomly selected from the lot. No more than one envelope shall be taken from any one unit package. For the lot to be acceptable there shall be no failure of any of the 10 envelopes tested to meet the specification requirements.

4.3.2.2 Envelopes with opaquing design. Envelopes having a printed opaquing design shall be tested as specified in 4.4.2 for compliance with the requirements in section 3. The lot shall consist of all envelopes having a printed opaquing design submitted for inspection at one time. Five envelopes for test shall be selected at random from the lot. For the lot to be acceptable, there shall be no failure of any of the five envelopes tested, to meet the specification requirements.

4.3.2.3 Envelopes having a temperature-humidity resistant adhesive. Envelopes having a temperature-humidity resistant flap adhesive shall be tested as specified in 4.4.3 to determine compliance with 3.11.2.2. The lot size shall be expressed in units of envelopes. The sample unit shall consist of five envelopes, randomly selected throughout the lot; not more than one envelope to be selected from any one package of envelopes. The inspection level for purposes of determining the sample size shall be S-2 and there shall be no failure of any envelope in the sample size to pass the test.

4.4 Test methods.

4.4.1 Adhesion of flap after moistening. Moisten the flap adhesive of each envelope in the sample size, using a moistener in accordance with GG-M-571 and seal the flap using moderate hand pressure. Fifteen seconds, -0, +5 seconds after sealing the envelope, raise the flap and note conformance with the requirements stated in 3.11.2 and 4.3.2.1.

4.4.2 Opacity of opaquing design. A sheet of 7 pound (17 by 22 - 500 sheets) manifold paper, (unglazed), bearing 12 point size, typewritten matter, shall be folded with the typewritten material to the outside and inserted in the envelope. The envelope and contents shall then be held between the observer and a 150 watt clear glass incandescent light source and at a distance of not more than 3 inches from the filament. No letter or figure shall be distinguishable.

4.4.3 Blocking of envelopes having temperature-humidity resistant flap adhesive. Prior to testing, the envelopes shall be conditioned for at least 24 hours at a temperature of 73° ± 3.5°F. and 50 ± 2 percent relative humidity. Stack five envelopes face down on a smooth flat surface with the flaps in a closed position and

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one above the other. Cover with a wooden block or metal sheet having a total weight, equivalent to 10 grams per square inch of envelope area. A weight tolerance of ± 5 percent shall be permitted. Place the envelopes and weight assembly in an oven maintained at 100 ± 2°F. and 80 ± 2 percent relative humidity for 24 hours. After 24 hours, remove the envelopes from the oven and allow to come to room temperature (73° ± 3.5°F.) with the weight removed. Examine each envelope flap for adhesion to the body of the envelope. To be considered satisfactory, the flap shall lift easily without disturbing the adhesive.

4.4.4 Writing quality. Write or draw several lines of words or characters on both sides of the paper using a pen with a firm medium point (Esterbrook point No. 9668 or equal) and a 1 percent aqueous solution of either C.I. Acid Red 1 (Color Index Number 18050) or C.I. Acid Green 3 (Color Index Number 42085) (see 6.4). To be considered satisfactory, the lines or characters shall be clearcut and legible without feathering or spreading.

4.4.5 Window material, opacity and reflectance.

4.4.5.1 Test equipment. Bausch and Lomb Spectronic 505 or equivalent.

4.4.5.2 Calibration. Prepare magnesium oxide reference standard in accordance with TAPPI method T 633. Calibrate the instrument according to the manufacturers instructions, except the reflectance at 100 percent shall be adjusted using the magnesium oxide as the reference standard.

4.4.5.3 Reflectance. Measure the reflectance of one sheet of window material, of the size specified in the instructions for the instrument being used, backed with a magnesium oxide standard at 400 millimicrons. The reflectance shall be made with all specular traps removed. The reflectance, in percent with respect to the reference curve at 400 millicrons shall be reported as the area reflectance (R_w).

4.4.5.4 Opacity. Repeat the determination as specified in 4.4.5.3 except with the sample backed by a cavity lined with black velvet or other material which will cause the reflectance of the cavity to be less than 1 percent, relative to magnesium oxide. The reflectance percent, with respect to the reference curve at 400 millimicrons, describes R_b. The opacity of the window material, in percent, is equal to R_b divided by R_w multiplied by 100.

4.4.6 Presence of optical brighteners.

4.4.6.1 Test specimens. Test two sheets from each test unit. The test area is a circle 5.7 cm in diameter. Make measurements at the top and bottom of both sides of each test sheet.

4.4.6.2 Equipment. Use a filter reflectometer with 0°-45° geometry, a light source with a color temperature of 3100°K, a blue filter approximating the CIE Z function, a movable ultraviolet-absorbing filter permitting inclusion or exclusion of the ultraviolet component from the light falling on the test area, and a non-fluorescent ceramic tile calibration standard with a known value for the function Z.

4.4.6.3 Procedure. (1) With the calibration standard positioned at the specimen window, adjust the instrument to give the reflectance value assigned to the standard. (2) With the unknown test specimen at the window and the ultraviolet filter positioned to include the ultraviolet component of the incident light, rotate the digital dial until the galvanometer shows zero deflection, and read the reflectance value from the dial. (3) Without moving the test specimen, and with the filter positioned to exclude the ultraviolet component, again rotate the digital dial to zero deflection and read the reflectance value. (4) Calculate the difference between readings (2) and (3); this is ΔB.

4.4.6.4 Report. Report ΔB for both wire and felt sides separately, and the average for both sides.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Envelope sizes up to and including 4-1/2 by 10-3/8 inches. Five hundred (500) envelopes of like description in sizes up to and including 4-1/2 by 10-3/8 inches shall be packaged in a close-fitting box conforming to PPP-B-676, type II, variety 2, class B, style 4. The box shall be closed in accordance with the appendix of the box specification. Sealing flaps of open side envelopes shall be folded down. Flaps of open end envelopes shall be folded down or scored and left extended. When specified (see 6.2), the sealing flaps of open side envelopes shall be scored and left extended to facilitate their use with automatic inserting machines. The envelopes shall be unbanded. When specified (see 6.2), they shall be separated in lots of 50 or 100 with separation slips or by turning up the flap of the fiftieth or one hundredth envelope. The separation slips shall be long enough to reach within one-half inch of the bottom of the box and to extend at least one inch above the envelopes.

5.1.1.2 Envelope sizes larger than 4-1/2 by 10-3/8 inches up to and including 9-1/2 by 12 inches. One thousand (1000) envelopes of like description in sizes larger than 4-1/2 by 10-3/8 up to and including 9-1/2 by 12 inches shall be packaged as specified in 5.1.1.1 except that the box shall conform to PPP-B-636, grade V2s. The box shall be closed, waterproofed, sealed and strapped, in accordance with the appendix to the box specification.

5.1.1.3 Envelope sizes larger than 9-1/2 by 12 inches. Five hundred (500) envelopes of like description in sizes larger than 9-1/2 by 12 inches shall be packaged as specified in 5.1.1.2.

5.1.2 Level B.

5.1.2.1 Envelope sizes up to and including 4-1/2 by 10-3/8 inches. Five hundred (500) envelopes of like description in sizes up to and including 4-1/2 by 10-3/8 inches shall be packaged as specified in 5.1.1.1 except that the box shall conform to PPP-B-676, type II, variety 1, class A or B. The box shall be closed in accordance with the appendix of the box specification.

5.1.2.2 Envelope sizes larger than 4-1/2 by 10-3/8 inches up to and including 9-1/2 by 12 inches. One thousand (1000) envelopes of like description in sizes larger than 4-1/2 by 10-3/8 inches up to and including 9-1/2 by 12 inches shall be packaged as specified in 5.1.1.1 except that the box shall conform to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.1.2.3 Envelope sizes larger than 9-1/2 by 12 inches. Five hundred (500) envelopes of like description in sizes larger than 9-1/2 by 12 inches shall be packaged as specified in 5.1.2.2.

5.1.3 Level C. The envelopes shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving agency.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Envelope sizes up to and including 2-1/2 by 4-1/4 inches. Twenty (20) boxes, packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, V2s. The box shall be closed, waterproofed, sealed, and strapped in accordance with the appendix to the box specification.

5.2.1.2 Envelope sizes larger than 2-1/2 by 4-1/2 inches up to and including 4-1/8 by 9-1/2 inches. Ten (10) boxes, packaged as specified in 5.1, shall be packed in a box specified in 5.2.1.1.

5.2.1.3 Envelope sizes larger than 4-1/8 by 9-1/2 inches up to and including 4-1/2 by 10-3/8 inches. Five (5) boxes, packaged as specified in 5.1, shall be packed in a box specified in 5.2.1.1.

5.2.1.4 Envelope sizes larger than 4-1/2 by 10-3/8 inches. Envelope sizes larger than 4-1/2 by 10-3/8 inches do not require additional packing.

5.2.2 Level B.

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5.2.2.1 Envelope sizes up to and including 2-1/2 by 4-1/4 inches. Twenty (20) boxes, packaged as specified in 5.1, shall be packed in a box conforming to PPP-B-636, class domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.2.2 Envelope sizes larger than 2-1/2 by 4-1/4 inches up to and including 4-1/8 by 9-1/2 inches. Ten (10) boxes, packaged as specified in 5.1, shall be packed in a box specified in 5.2.2.1.

5.2.2.3 Envelope sizes larger than 4-1/8 by 9-1/2 inches up to and including 4-1/2 by 10-3/8 inches. Five (5) boxes, packaged as specified in 5.1, shall be packed in a box specified in 5.2.2.1.

5.2.2.4 Envelope sizes larger than 4-1/2 by 10-3/8 inches. Envelope sizes larger than 4-1/2 by 10-3/8 inches do not require additional packing.

5.2.2.5 When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification.

5.2.3 Level C. The envelopes shall be packed to assure carrier acceptance and safe delivery to destination. Containers shall comply with the requirements of the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable to the mode of transportation.

5.3 Marking. Marking shall be in accordance with 5.3.1 or 5.3.2 as specified (see 6.2).

5.3.1 Civil agencies. In addition to markings required by the contract or order, the packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military requirements. In addition to markings required by the contract or order, the packages and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Mailing envelopes furnished under this specification are intended to cover the needs of the following primary users: All departments and independent establishments and wholly owned Government corporations in the executive branch of the Federal Government, within the 50 States, and the District of Columbia, and the Commonwealth of Puerto Rico.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, form (see 1.2.1).
(c) Size (see 3.4).
(d) Color (see 3.7.1).
(e) Weight of type I (see table I).
(f) Weight of types II and III (see table II).
(g) Corners, size, and location of window when window envelopes are required (see 3.10.2).
(h) When temperature-humidity resistant flap adhesive is required (see 3.11.2.2).
(i) Whether ungummed or split gummed flaps are required (see 3.12.5.2 and 3.12.5.3).
(j) Printing, ink, type, location of borders, marking, and design (see 3.13 through 3.13.5).
(k) Make and model of inserting machine, as applicable (see 3.12.5.1).
(l) Level of packaging and packing required (see 5.1 and 5.2).
(m) Whether envelopes are to be packaged with flaps extended (see 5.1.1.1, 5.1.1.2, 5.1.1.3, 5.1.2.1, 5.1.2.2 and 5.1.2.3).
(n) Whether each 50 or 100 envelopes are to be separated (see 5.1.1.1, 5.1.1.2, 5.1.1.3, 5.1.2.1. 5.1.2.2, and 5.1.2.3).
(o) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.5).
(p) Whether marking shall be in accordance with civil agencies or military requirements (see 5.3).

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6.3 Standard sample. Standard sample for color of light brown, light green, and blue paper may be obtained upon application to the procuring agency issuing the invitation for bids (see 3.5.1 and 3.7.1.1).

6.4 Color designations are those listed in the Technical Manual of the American Association of Textile Chemists and Colorists. Acid Red 1, Color Index Number 18050 may be obtained from Textile Aniline and Chemical Company. Acid Green 3, Color Index Number 42085 may be obtained from the following: Sandoz, Incorporated; National Aniline Division of Allied Chemical Corporation; Geigy Chemical Corporation, and American Cyanamid Corporation.

MILITARY INTEREST:

Custodians:

Army-GL
Navy-SA
Air Force - 84

User activities:

Army-EL
Navy-SH, YD, CG, MC

Preparing activity:

GSA-FSS

CIVIL AGENCY COORDINATING ACTIVITIES:

AGR - SCS
AEC
COM - NBS
GSA - FSS
HEW - SSA
INT - MIN
JUS - FPI
NASA - JFK
PO
TVA
TR - OAP
VA - DMS
HUD - HHE

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HIGH-CUT DIAGONAL SEAM

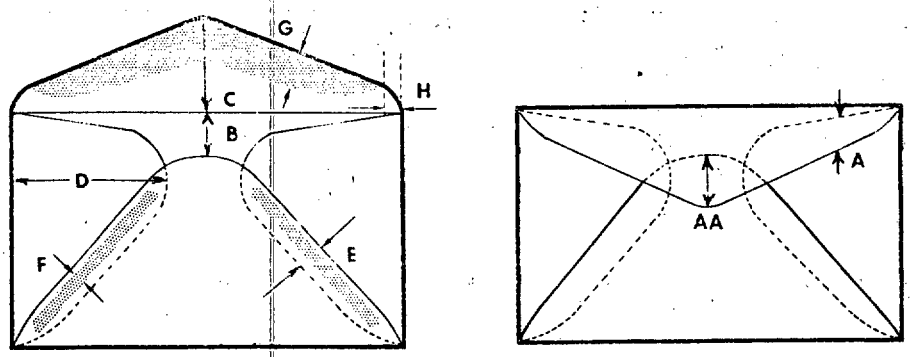


Illustration No. 1

WEB-FED POINT FLAP

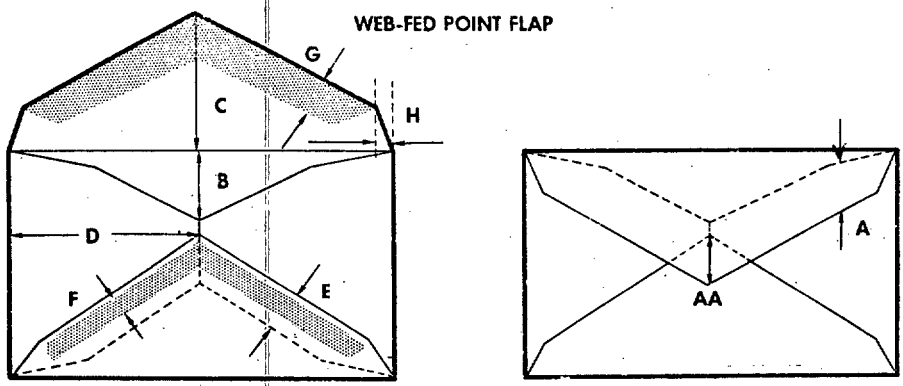


Illustration No. 2

SIDE SEAM

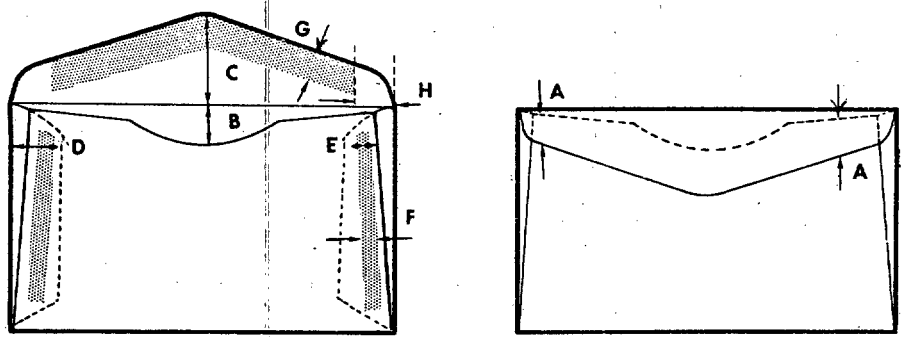


Illustration No. 3

CENTER SEAM

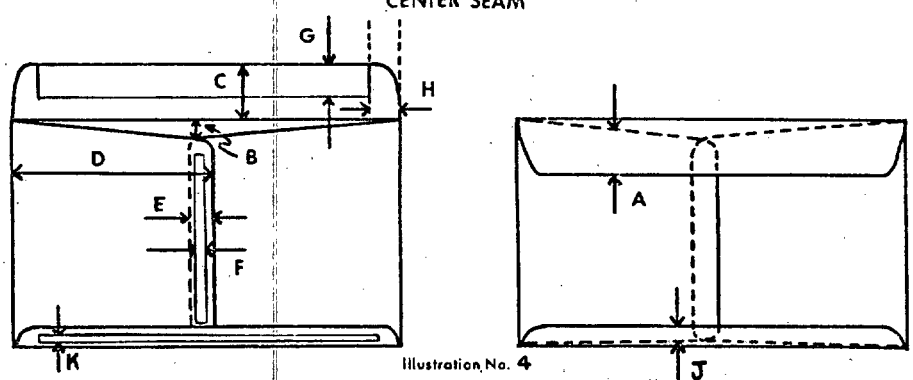


Illustration No. 4

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